

scVI Human Validation — Reviewer 1 (Blinded)

Purpose

Independently determine how each citing paper uses scVI. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Calibration: complete CAL-01 to CAL-03 independently, then discuss those three cases and freeze the codebook.

Formal validation: complete every HV row in the displayed order. Do not open the other reviewer document or the consensus document until both formal reviews are locked.

Reconciliation: after both reviewers lock their answers, the coordinator copies both records into the consensus document. Discuss only disagreements. Use `REFERENCE_UNRESOLVED` when agreement is not possible.

Evidence boundary: use only the linked approved article versions. Search all occurrences of “scVI” and inspect the surrounding methods, results, figures, tables, and supplements that are linked.

Allowed values

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Use labels (select all that apply): Benchmark / Extend or develop / Apply to biological data / Other

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located

Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

Review table

Article and approved evidence links	Reviewer 1 independent record
CAL-01 CALIBRATION Cell Subtype Classification via Representation Learning Based on a Denoising Autoencoder for Single-Cell RNA Sequencing 2021 IEEE Access DOI: 10.1109/access.2021.3052923 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: To perform fundamental analysis for single-cell transcriptome dataset, the single-cell variational inference (scVI) model was introduced for probabilistic representation [34] and ACTINN implemented a DNN model to assign each cell a cell type [35]. Notes: Reviewer initials: SM Review completed: Yes
CAL-02 CALIBRATION Eleven grand challenges in single-cell data science 2020 Genome biology DOI: 10.1186/s13059-020-1926-6 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Some of them are expressly aimed at imputation (e.g., AutoImpute, DeepImpute, EnImpute, DCA, and scVI). Notes: Reviewer initials: SM Review completed: Yes

CAL-03 CALIBRATION Mosaic integration of spatial multi-omics with SpaMosaic 2024 DOI: 10.1101/2024.10.02.616189 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:For RNA-seq data, dimensionality reduction and batch effect correction were performed using scVI with the zero-inflated negative binomial likelihood. Notes: Reviewer initials:SM Review completed:Yes
HV-005 FORMAL VALIDATION Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer. 2026 DOI: 10.1016/j.ccell.2025.12.003 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:The scVI model was trained on the 6000 most highly variable genes as determined with scanpy's¹¹⁰ <i>pp.highly_variable_genes</i> with parameters <i>flavor="seurat_v3"</i> and <i>batch_key="dataset"</i>. Notes: Reviewer initials:SM Review completed:Yes

HV-048 FORMAL VALIDATION Beyond the List: Bioagent-Agnostic Signatures Could Enable a More Flexible and Resilient Biodefense Posture Than an Approach Based on Priority Agent Lists Alone 2021 Pathogens DOI: 10.3390/pathogens10111497 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Scientists also need better models to eliminate nuisance technical variation that arises between different runs of the same single-modality 'omics experiment (i.e., batch effects) [107,108]. Notes: Reviewer initials:SM Review completed:Yes
HV-027 FORMAL VALIDATION Multi-modal Diffusion Model with Dual-Cross-Attention for Multi-Omics Data Generation and Translation 2025 DOI: 10.1101/2025.02.27.640020 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:For comparison, we also generated multi-omics data using other simulators, including MultiVI¹⁶, CFGen¹⁷, and scDesign3¹⁸, as well as two single-modal generators: scVI¹⁹ and peakVI²⁰, which generate scRNA-seq and scATAC-seq data, respectively. Notes: Reviewer initials:SM Review completed:Yes

HV-043 FORMAL VALIDATION Benchmarking atlas-level data integration in single-cell genomics 2021 Nature Methods DOI: 10.1038/s41592-021-01336-8 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Overall, scANVI, Scanorama, scVI and scGen perform well, particularly on complex integration tasks, while single-cell ATAC-sequencing integration performance is strongly affected by choice of feature space. Notes: Reviewer initials:SM Review completed:Yes
HV-002 FORMAL VALIDATION Multimodal Hypersampling of the adult human heart resolves tissue-specific cell states and disease-associated variants 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.16.732617 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:A secondary latent representation was generated with scVI⁹¹, trained per cell type on the raw counts of the highly variable genes using the same composite batch_key as the batch covariate; total counts, the percentage of mitochondrial reads and the percentage of ribosomal reads were supplied as continuous covariates, and gene- and batch-specific dispersion was used. This model used a 30-dimensional latent space and a three-layer encoder (increased to 4 layers for the global dataset). This representation was used only to corroborate the robustness of the annotations. Notes: Reviewer initials:SM Review completed:Yes

HV-010 FORMAL VALIDATION Single-cell characterization of anti-LAG-3 and anti-PD-1 combination treatment in patients with melanoma 2023 Journal of Clinical Investigation DOI: 10.1172/jci164809 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:To overcome the batch effect, we used a probabilistic framework to account for different nuisance factors of variation in an unsupervised manner with a deep generative modeling method scVI. Notes: Reviewer initials:SM Review completed:Yes
HV-029 FORMAL VALIDATION Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design 2020 Molecules DOI: 10.3390/molecules25143250 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:It should be noted that the scVI and SAUCIE methods also employ deep learning approaches such as deep autoencoder algorithms, but not the GAN architecture. Notes: Reviewer initials:SM Review completed:Yes

HV-032 FORMAL VALIDATION Effectively Clustering Single Cell RNA Sequencing Data by Sparse Representation 2021 IEEE Transactions on Computational Biology and Bioinformatics DOI: 10.1109/tcbb.2021.3128576 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Lopez et al. [20] introduced a single-cell variational inference (scVI) to derive the probabilistic representations of scRNA-seq data by deep generative model variational auto-encoder. Notes: Reviewer initials:SM Review completed:Yes
HV-004 FORMAL VALIDATION Hybrid deep learning–mechanistic modeling of cellular dynamics from a spatiotemporal single-cell atlas 2025 DOI: 10.1101/2025.10.02.679951 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Deep generative models such as single-cell variational inference (scVI) (13) and continuous-time neural model (2) motivate our “VAE with latent neural ODE” design. Notes: Reviewer initials:SM Review completed:Yes

HV-025 FORMAL VALIDATION PI3Ky in B cells promotes antibody responses and generation of antibody-secreting cells 2024 Nature Immunology DOI: 10.1038/s41590-024-01890-1 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6 Supplement 7 Supplement 8 Supplement 9 Supplement 10	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:Thereafter, the filtered data were integrated and batch effects corrected using single-cell variational inference (scVI)⁵⁹. Notes: Reviewer initials:SM Review completed:Yes
HV-015 FORMAL VALIDATION Recovering Single-cell Heterogeneity Through Information-based Dimensionality Reduction 2021 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2021.01.19.427303 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:For comparison, we also tested PCA, ICA, scVI, and six rare cell type discovery tools: RaceID [8], GiniClust [9], CellSIUS [10], FiRE [11], GeoSketch [12], and Hopper [13]. Notes: Reviewer initials:SM Review completed:Yes

HV-028 FORMAL VALIDATION Deep learning applications in single-cell genomics and transcriptomics data analysis 2023 Biomedicine & Pharmacotherapy DOI: 10.1016/j.biopha.2023.115077 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Though such end-to-end DL models remain nascent, preliminary works in this area (such as the scVI framework [40]) have shown promising results, demonstrating that DL models hold a great potential in the development of end-to-end SC pipelines. Notes: Reviewer initials:SM Review completed:Yes
HV-030 FORMAL VALIDATION Epigenetic biomarkers in inflammatory bowel diseases—computational challenges and opportunities 2026 Journal of Crohn s and Colitis DOI: 10.1093/ecco-jcc/jjag022 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Probabilistic modeling approaches such as scVI and more classical tools like ComBat explicitly model batch-specific technical factors alongside true biological variability. Notes: Reviewer initials:SM Review completed:Yes

HV-045 FORMAL VALIDATION An overview of computational methods in single-cell transcriptomic cell type annotation 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf207 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:This strategy was later adopted by deep learning methods, such as single-cell variational inference (scVI) [100], where the encoder preferentially processed the HVG subset. Notes: Reviewer initials:SM Review completed:Yes
HV-035 FORMAL VALIDATION Sequencing dropout-and-batch effect normalization for single-cell mRNA profiles: a survey and comparative analysis 2020 Briefings in Bioinformatics DOI: 10.1093/bib/bbaa248 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Nonetheless, moderate difference still remains in both of the medium and high degrees of batch effect simulated datasets after normalization by scVI and still in the datasets with high degrees of batch effect after normalization by Seurat. Notes: Reviewer initials:SM Review completed:Yes

HV-046 FORMAL VALIDATION Projecting single-cell transcriptomics data onto a reference T cell atlas to interpret immune responses 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.06.23.166546 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Several computational methods have been developed to correct technical biases introduced by handling experiments in batches, and to align datasets over their biological similarities Notes: Reviewer initials:SM Review completed:Yes
HV-049 FORMAL VALIDATION Trends and Potential of Machine Learning and Deep Learning in Drug Study at Single-Cell Level 2023 Research DOI: 10.34133/research.0050 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:SC3 [99], DrImpute [101], SIMLR [98], Seurat V2 [100], AutoImpute [102], DCA [112], AutoClass [126], Lin's NN [6], DeepImpute [119], cTP-net [120], Monocle2 [97], scVI [103], scGen [20], VASC [108], scvis [109], scImpute [110], ZINB-WaVE [105], scScope [111], SAUCIE [115], SAVER-X [113], scGNN [125], SpaGCN [124], Reuben [122], MNN [107], Wang [114] Notes: Reviewer initials:SM Review completed:Yes

HV-042 FORMAL VALIDATION Pertpy: an end-to-end framework for perturbation analysis 2025 Nature Methods DOI: 10.1038/s41592-025-02909-7 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:As pertpy is integrated with the scverse ecosystem, users of pertpy can seamlessly integrate established batch correction methods Notes: Reviewer initials:SM Review completed:Yes
HV-023 FORMAL VALIDATION Transcriptomic advances in studies of muscle stem cell aging: From bulk to single-cell and beyond 2026 Cell Research DOI: 10.1038/s41422-026-01240-w Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:By contrast, scVI, a variational autoencoder framework widely used in single-cell analysis, has already been applied to datasets containing MuSCs, such as those from the Tabula Sapiens Project,^{68,69} and serves as a useful benchmark for modeling single-cell variation. Notes: Reviewer initials:SM Review completed:Yes

HV-020 FORMAL VALIDATION SCIntRuler: guiding the integration of multiple single-cell RNA-seq datasets with a novel statistical metric 2024 Bioinformatics DOI: 10.1093/bioinformatics/btae537 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Notable deep learning approaches include scVI (Lopez et al. 2018), trVAE (Lotfollahi et al. 2020), and scGen (Lotfollahi et al. 2019). Notes: Reviewer initials:SM Review completed:Yes
HV-040 FORMAL VALIDATION Best practices for single-cell analysis across modalities 2023 Nature Reviews Genetics DOI: 10.1038/s41576-023-00586-w Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:Depending on the complexity of the integration tasks, such as atlas integration, deep-learning approaches such as scANVI^{40}, scVI^{41} and scGen^{42} as well as linear-embedding models such as Scanorama^{43} performed best, whereas for less complex integration tasks, Harmony^{37} is the preferred method^{35}. Notes: Reviewer initials:SM Review completed:Yes

HV-047 FORMAL VALIDATION Deep learning in single-cell and spatial transcriptomics data analysis: advances and challenges from a data science perspective 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf136 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:An example is scVI [105], which explicitly incorporates batch annotations and addresses batch effects through conditional independence assumptions (Fig. 4d). Notes: Reviewer initials:SM Review completed:Yes
HV-038 FORMAL VALIDATION Deep-learning methods for unveiling large-scale single-cell transcriptomes 2024 Cancer Biology and Medicine DOI: 10.20892/j.issn.2095-3941.2023.0436 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Single-cell variational inference (scVI) 5 takes batch information into a VAE for batch correction. S Notes: Reviewer initials:SM Review completed:Yes

HV-017 FORMAL VALIDATION Multi-omic identification of perineurial hyperplasia and lipid-associated nerve macrophages in human polyneuropathies 2025 Nature Communications DOI: 10.1038/s41467-025-62964-8 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:Batch effects were accounted for by integrating the samples with scVI v1.0.4⁵⁴. Notes: Reviewer initials:SM Review completed:Yes
HV-050 FORMAL VALIDATION Batch correcting single-cell spatial transcriptomics count data with Crescendo improves visualization and detection of spatial gene patterns 2025 Genome biology DOI: 10.1186/s13059-025-03479-9 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We next compared Crescendo's ability to batch correct individual genes in this mouse brain spatial dataset to five representative state-of-the-art algorithms: ComBat-Seq, scVI, Seurat anchor integration, MNN, and limma (" Methods"). Notes: Reviewer initials:SM Review completed:Yes

HV-018 FORMAL VALIDATION Hierarchical classification of immune cell transcriptomes at population-scale 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.05.30.728980 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:We supplied the dataset key as covariate key, ran SCVI (with 2 layers in encoder and decoder, covariate encoding set to True, covariate deep injection set to True, layer normalization for encoder and decoder and without batch normalization) for a maximum of 80 epochs. Notes: Reviewer initials:SM Review completed:Yes
HV-013 FORMAL VALIDATION COVID-Datathon: Biomarker identification for COVID-19 severity based on BALF scRNA-seq data 2021 arXiv (Cornell University) DOI: 10.48550/arxiv.2110.04986 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:As a result, it may not be beneficial to apply typical scRNA-seq tools for biomarker identification and cell clustering on this normalized data set, such as scVI [Lopez et al., 2018], DESeq2 [Love et al., 2014] and SimCD [Niyakan et al., 2021], in which Negative Binomial (NB) distribution models and their extensions were developed to model scRNA-seq data Notes: Reviewer initials:SM Review completed:Yes

HV-044 | FORMAL VALIDATION

Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN

2024 | Nature Methods

DOI: 10.1038/s41592-024-02191-z

[Article page](#)

[Full text 1](#)

[Full text 2](#)

Review status: LOCKED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Benchmark

Biological insight: No

Evidence sufficiency: Sufficient

Supporting quote + locator: We compare the performance of SATURN to another single-cell multispecies integration method, SAMap¹¹, and unsupervised integration methods Harmony⁶, scVI⁸ and Scanorama⁷.

Notes:

Reviewer initials: SM

Review completed: Yes

HV-022 | FORMAL VALIDATION

Transforming microfluidics for single-cell analysis with robotics and artificial intelligence

2025 | Lab on a Chip

DOI: 10.1039/d5lc00216h

[Article page](#)

[Full text 1](#)

Review status: LOCKED

Citation validity: Yes

Method executed: No

Use labels (select all that apply):

Primary use: Mention only

Biological insight: No

Evidence sufficiency: Sufficient

Supporting quote + locator: For example, adaptive sampling algorithms can guide iterative single-cell RNA sequencing to focus on rare or transitional cell states. In the realm of data integration, cutting-edge AI models such as variational autoencoders (e.g., scVI¹⁶³ and TotalVI¹⁶⁴) and graph neural networks^{165,166} have proven effective at aligning and integrating diverse modalities including transcriptomics, proteomics, and epigenomics, enabling comprehensive characterization of cellular states across multiple molecular layers.

Notes:

Reviewer initials: SM

Review completed: Yes

HV-041 FORMAL VALIDATION Decoding disease and therapy through multiomics integration and systems analysis 2026 Briefings in Bioinformatics DOI: 10.1093/bib/bbag311 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Similarly, scVI (single-cell variational inference) employs deep generative modelling with variational autoencoders to represent gene expression distributions, allowing for probabilistic batch correction and imputation [36]. Notes: Reviewer initials:SM Review completed:Yes
HV-039 FORMAL VALIDATION SCOPE: Localizing fate-decision states and their regulatory drivers in single-cell differentiation 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.04.07.717037 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed: Use labels (select all that apply):Apply to biological data/Extend Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:Let $X \in \mathbb{R}^{n \times p}$ be a dataset with n cells and p highly variable genes, where X_{ij} denotes the denoised library-size-adjusted gene expression of gene j in cell i. Broadly speaking, we use Scanpy [59] to determine the highly variable genes, and use scVI [60] or peakVI [61] to normalize the count matrix of RNA or ATAC, respectively, whenever possible. If the count matrix is unavailable, a log-normalized matrix can be used as input. Notes: Reviewer initials:SM Review completed:Yes

HV-031 FORMAL VALIDATION Candida albicans exhibits distinct cytoprotective responses to anti-fungal drugs that facilitate the evolution of drug resistance 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.01.21.914549 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:We removed genes that expressed in less than 20 cells. Normalization, imputation and batch correction were performed by scVI^{58,59}, a tool which implements a probabilistic model of mRNA capture and uses a variational autoencoder to estimate priors across batches and conditions (Supplemental Figure 1). Notes: Reviewer initials:SM Review completed:Yes
HV-036 FORMAL VALIDATION SECANT: a biology-guided semi-supervised method for clustering, classification, and annotation of single-cell multi-omics 2022 PNAS Nexus DOI: 10.1093/pnasnexus/pgac165 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:General workflow of SECANT. In this study, we used manual gating or Gaussian mixture model (GMM) to classify confident cell types with ADT data, and used scVI for dimensional reduction and batch effect correction with RNA data for data preprocessing. Notes: Reviewer initials:SM Review completed:Yes

HV-008 FORMAL VALIDATION LAIOR: a hyperbolic neural ODE variational framework for interpretable single-cell manifold learning and trajectory inference 2026 Frontiers in Genetics DOI: 10.3389/fgene.2026.1838613 Article page Full text 1 Supplement 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We compared LAIOR with 23 baseline methods across six paradigms: classical methods (PCA, ICA, factor analysis), generative models (scVI, scGNN, scDiffusion, siVAE), contrastive learning (scGCC, CLEAR), hyperbolic embeddings (Poincaré maps, PGM, learnable Poincaré Gaussian mixture (LPGM), hyperbolic wrapped normal (HWN), scDHMAP), factor-based VAEs (β-VAE, InfoVAE, TC-VAE, DIP-VAE), and modality-specific methods (PeakVI, PoissonVI, scTour). Notes: Reviewer initials:SM Review completed:Yes
HV-034 FORMAL VALIDATION Single-Cell Multi-Omics: Insights into Therapeutic Innovations to Advance Treatment in Cancer 2025 International Journal of Molecular Sciences DOI: 10.3390/ijms26062447 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Several horizontal integration tools have been developed for single-cell datasets, including Seurat, Harmony, LIGER, and scVI Notes: Reviewer initials:SM Review completed:Yes

HV-014 FORMAL VALIDATION Kinetic Foundation of the Zero-Inflated Negative Binomial Model for Single-Cell RNA Sequencing Data 2020 SIAM Journal on Applied Mathematics DOI: 10.1137/19m1253198 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity: Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Among these statistical models, the zero-inflated negative binomial (ZINB) model is the most widely used [20–31], where the zero-inflated part describes dropouts and the negative binomial part accounts for over-dispersion. Some other commonly used models are listed in Sec. 7. Notes: Reviewer initials:SM Review completed:Yes
HV-016 FORMAL VALIDATION AtlasLens: Metadata-centric exploration and analysis of single-cell atlases 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.10.735823 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:AtlasLens works with both single-cell dataset and atlases containing rich, diverse metadata built using integration tools such as Seurat and scVI (Lopez et al., 2018). To use AtlasLens and its functions (step 4), integrated atlases should be saved as a Seurat object; conversion scripts for AnnData and related formats are provided in the repository. Notes: Reviewer initials:SM Review completed:Yes

HV-011 FORMAL VALIDATION JIND-Multi: Leveraging Multiple Labeled Datasets for Automated Annotation of Single-Cell RNA and ATAC Data 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.15.633130 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Various integration methods have been proven effective, such as Harmony [3] and scVI [4] for scRNA-Seq, while Seurat [5] and LIGER [6] can be used for both scRNA-Seq and single-cell ATAC sequencing (scATAC-Seq) data. Notes: Reviewer initials:SM Review completed:Yes
HV-024 FORMAL VALIDATION A multi-dataset single-cell meta-analysis of human keloid skin across Asian, Black and White populations reveals endothelial and mesenchymal programs linked to ethnic disparities 2026 Frontiers in Medicine DOI: 10.3389/fmed.2026.1751725 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:After quality control and batch integration (scVI/scANVI), we annotated cell types and subtypes. Notes: Reviewer initials:SM Review completed:Yes

HV-019 FORMAL VALIDATION Anti-bias training for (sc)RNA-seq: experimental and computational approaches to improve precision 2021 Briefings in Bioinformatics DOI: 10.1093/bib/bbab148 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For instance, dimension reduction based on zero-inflated distributions was used on scRNA-seq datasets to extract informative variables for further analysis (e.g. ZIFA [105], ZINB-WaVE [106] or ScVI [107]). Notes: Reviewer initials: SM Review completed: Yes
HV-021 FORMAL VALIDATION Biologically Informed Variational Inference Enables Interpretable Cell Phenotyping and Discovery 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.06.10.657924 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: NetworkVI builds upon the multiVI model (Ashuach et al. 2023) and thus leverages modality-specific decoders; scVI for transcriptome (Lopez et al. 2018), PeakVI for chromatin accessibility (Ashuach et al. 2022), and totalVI for epitope data (Gayoso et al. 2021). Notes: Reviewer initials: SM Review completed: Yes

HV-009 FORMAL VALIDATION Coding agents author interpretable single-cell embedding models from the literature 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.07.737048 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Specifically, the blueprint (both grounding variants), UCell and expiMap form one tight cluster high on the batch axis, well above PCA, scVI-census, UCE and scVI without its batch key. Notes: Reviewer initials:SM Review completed:Yes
HV-006 FORMAL VALIDATION Implementing distinct spatial proteogenomic technologies: opportunities, challenges, and key considerations 2024 Clinical & Experimental Immunology DOI: 10.1093/cei/uxae077 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Some widely used examples are Harmony [68], which excels with smaller datasets and scVI [69], which is more appropriate for complex integration tasks [70, 71]. Notes: Reviewer initials:SM Review completed:Yes

HV-033 FORMAL VALIDATION HoloCell: A Generative Foundation Model for Holistic Cellular Modeling 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.07.730684 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: In the fine-tuned setting, comparisons were made against scVI for transcriptomics, scBasset for epigenomics, and cyCombine for proteomics. Notes: Reviewer initials:SM Review completed:Yes
HV-003 FORMAL VALIDATION BrainBridge Characterizes Key Factors affecting Alzheimer's Disease and Associated Phenotypes 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2025.12.03.692255 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Use labels (select all that apply):Benchmark/Extend Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Furthermore, we also include MrVI [25] as a new baseline method. MrVI is a computational tool developed based on scVI [39] to model sample-level covariates. Notes: Reviewer initials:SM Review completed:Yes

HV-001 FORMAL VALIDATION Computational tools for analyzing single-cell data in pluripotent cell differentiation studies 2021 Cell Reports Methods DOI: 10.1016/j.crmeth.2021.100087 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Although some of these methods rely on linear transformations of one dataset onto the other, based on identified anchor cells (Alavi and Bar-Joseph, 2020; Haghverdi et al., 2018; Stuart et al., 2019), others utilize neural network models to embed both datasets in a shared reduced dimension (Johansen and Quon, 2019; Lopez et al., 2018). Notes: Reviewer initials:SM Review completed:Yes
HV-012 FORMAL VALIDATION sCellIST: a Multiple Instance Learning approach to predict single-cell gene expression from H&E images using spatial transcriptomics 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.11.07.622225 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:$p_{NB}(\cdot; a, b)$: density function of a negative binomial distribution parametrized by the parameters a and b corresponding to either μ (mean) and θ (inverse overdispersion) used in ref. 48 or total counts and probability of success parametrisation. Notes: Reviewer initials:SM Review completed:Yes

HV-026 FORMAL VALIDATION Tumor-infiltrating CD27 - IgD - regulatory B cells suppress cytotoxic CD8 + T cell responses in renal cell carcinoma 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.07.08.663720 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Use labels (select all that apply): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:To preserve cancer-driven clusters, we applied single-cell Variational Inference (scVI) to separately integrate and cluster the available single-cell B cell datasets for each cancer type (Table S1A, Figure S1A for schematic describing the pipeline). Notes: Reviewer initials:SM Review completed:Yes
HV-007 FORMAL VALIDATION Biological network-inspired interpretable variational autoencoder 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.12.17.423310 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Use labels (select all that apply): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:More recently, deep generative models such as Variational Autoencoders⁷ (VAEs) have proven to be extremely useful for the probabilistic modeling of single-cell transcriptomes, such scVI or scGen ⁸ Notes: Reviewer initials:SM Review completed:Yes

HV-037 | FORMAL VALIDATION

**Assessing Markovian and Delay Models for Single-Nucleus
RNA Sequencing**

2023 | Bulletin of Mathematical Biology

DOI: 10.1007/s11538-023-01213-9

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Review status: NOT STARTED

Citation validity:No

Method executed:No

Use labels (select all that apply):

Primary use:Mention only

Biological insight:No

Evidence sufficiency:Insufficient

Supporting quote + locator:

Notes:

Reviewer initials:SM

Review completed:Yes